

Radiofrequency electrical conductivity reveals a distinct dimension of early human brain development

Arnaud Boutillon

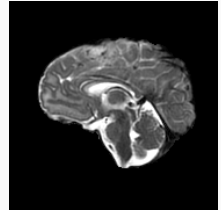
July 17, 2026

CDB Seminar

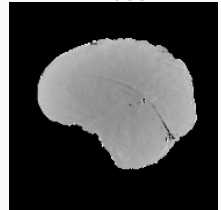
What else is hiding in MRI data?

- MRI scanner produces magnitude and phase images.
- Phase encodes the local **radiofrequency electrical conductivity** of tissue.

Magnitude



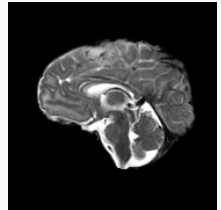
Phase



What else is hiding in MRI data?

- Phase data is routinely discarded.
- MRI studies have been rejecting a **biophysical measurement** of the brain.

Magnitude



The perinatal brain changes faster than at any other stage of life

Morphological atlas

- Early human brain development is characterised by rapid and large-scale **change in tissue composition and architecture.**

Schuh et al., 2018

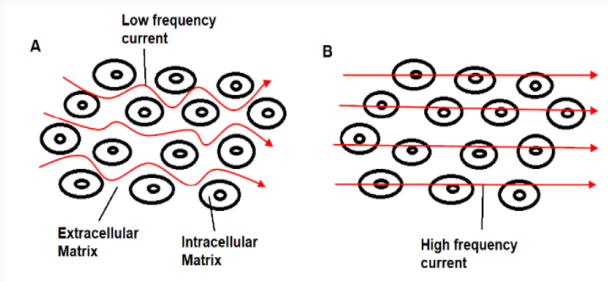
The perinatal brain changes faster than at any other stage of life

- Diffusion MRI probes water mobility and is sensitive to **microstructural maturation**.
- Radiofrequency electrical conductivity tracks **ionic composition and electronic environment** (Pethig et al., 1984)
→ **a distinct biophysical dimension**.

Morphological atlas

Schuh et al., 2018

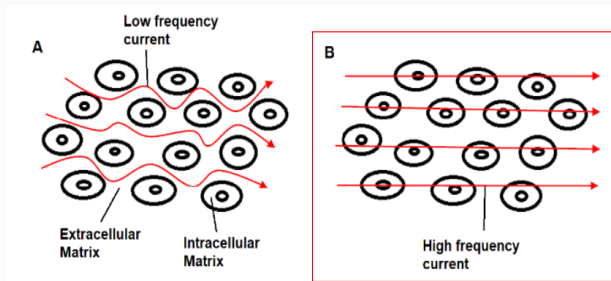
What is electrical conductivity?



Amini et al., 2018

- Conductivity reflects how **electrical current flows through tissue**.
- Current exhibits a strong frequency dependence.
 - At low-frequency → ionic conduction.
 - At high-frequency → **wave propagation**.

What is electrical conductivity?



Amini et al., 2018

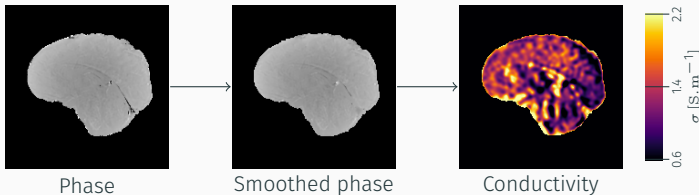
- At 128 MHz (i.e. 3T), conductivity is characterised by:
 - Water content and free ions concentration/mobility.
 - Membrane interface density.
 - Extracellular space volume fraction.

How do we measure radiofrequency electrical conductivity?

- Conductivity reconstruction involves solving a **simplified form** of Maxwell's equations from the phase distribution.
- It requires:
 - no additional hardware,
 - **no dedicated sequences.**
- Radiofrequency electrical conductivity can be extracted from the **phase of standard MRI scans** (TSE, GRE, bSSFP, EPI, etc.).

Reconstruction pipeline

- Typical conductivity reconstruction pipelines involve:
 - Image pre-processing including **smoothing**.
 - **Reconstruction algorithm** to estimate the conductivity map.
 - Extraction of **conductivity averages** within tissue compartments.



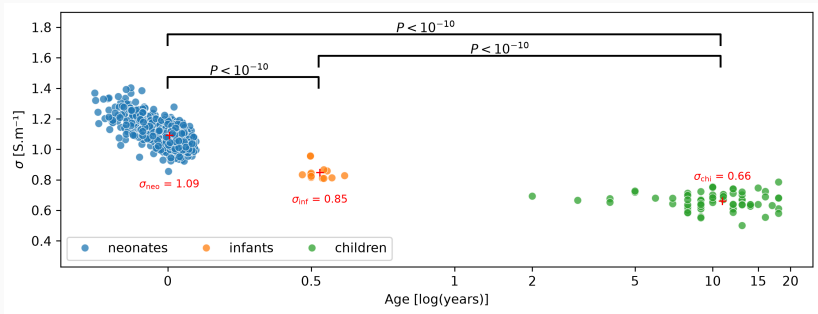
- We analysed **1015 scans spanning 26 weeks PMA to 18 years**, from four independent cohorts:

Cohort	<i>n</i>	Age
Neonates (dHCP)	888	26–45 weeks PMA
HIE neonates	35	37–47 weeks PMA
Infants (BIBS)	14	~6 months
Children (RICE)	78	2–18 years

- We analysed **1015 scans spanning 26 weeks PMA to 18 years**, from four independent cohorts and we investigated:
 - How does conductivity change across **early development**?
 - Is conductivity sensitive to **prematurity**?
 - Does conductivity capture something different from diffusion MRI in **brain injury**?

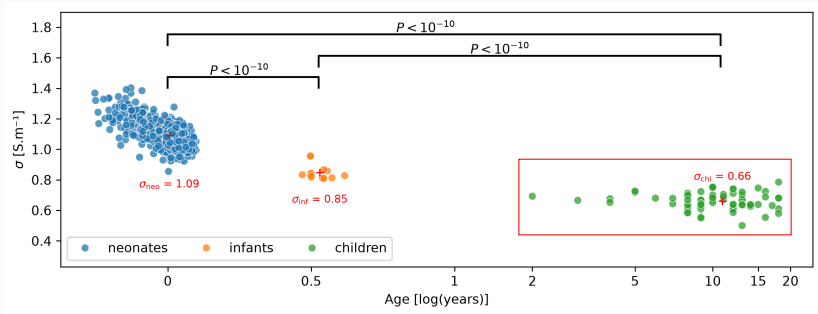
Developmental model

Conductivity declines across early human development



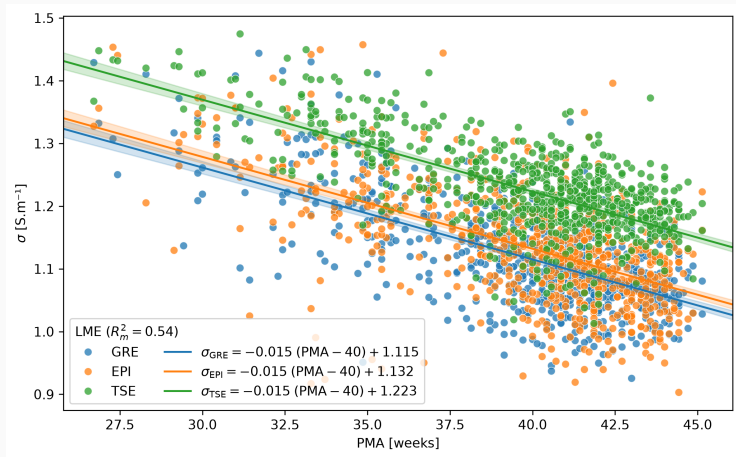
- Per-subject **whole-brain parenchymal** electrical conductivity.
- **Progressive decline** in conductivity across early life.
- Apparent **stabilisation** after two years.

Conductivity declines across early human development



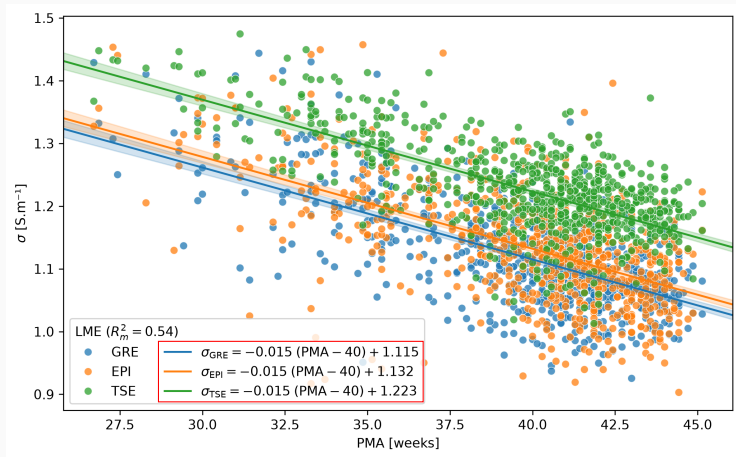
- Conductivity in children brains within range of **adult literature values** (Gurler and Ider, 2017, He et al., 2024)

Conductivity decline is consistent across independent methods



- Neonatal conductivity reconstructed from three independent MRI sequences: GRE, EPI and TSE.
- **Developmental slopes** were concordant across methods.

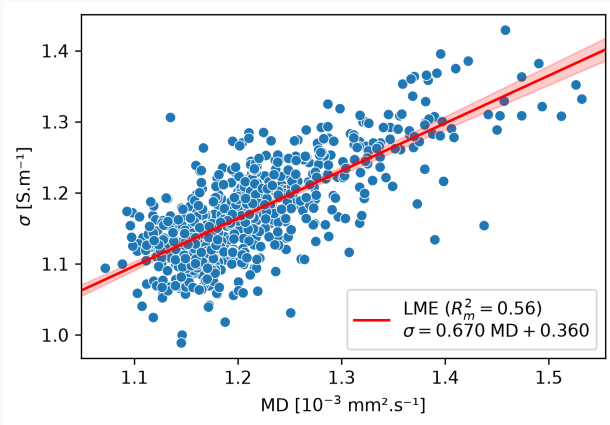
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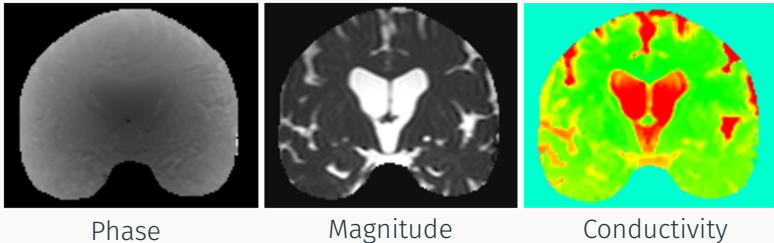
Developmental model with correction factors

Conductivity and MD association



- Decrease in conductivity could be explained by **reduction in water content**.
- Strong MD-conductivity correlation due to **shared dependence on water content**.

Contamination from ventricular CSF



Katscher et al., 2017

- Ventricular CSF conductivity exceeds that of brain parenchyma.
- **Contamination** via partial volume effects (PVEs).
- Smoothing mix CSF signal with tissue conductivity.

Developmental model correction factors

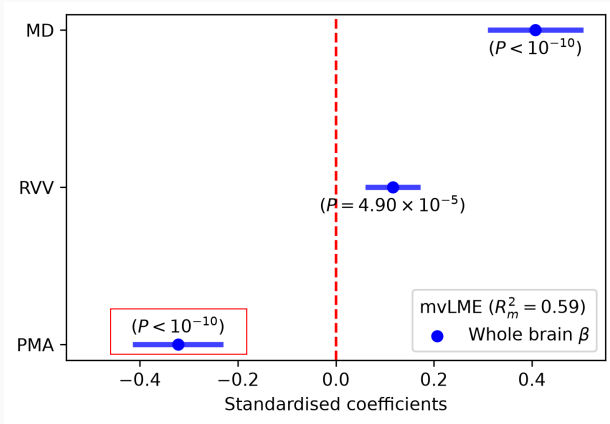
- Lack of direct water content measurement
→ **MD used as proxy.**
- Correction of PVEs and spatial smoothing
→ Relative Ventricular Volume (RVV) used as surrogate.

$$\text{Conductivity} \sim \text{PMA}$$



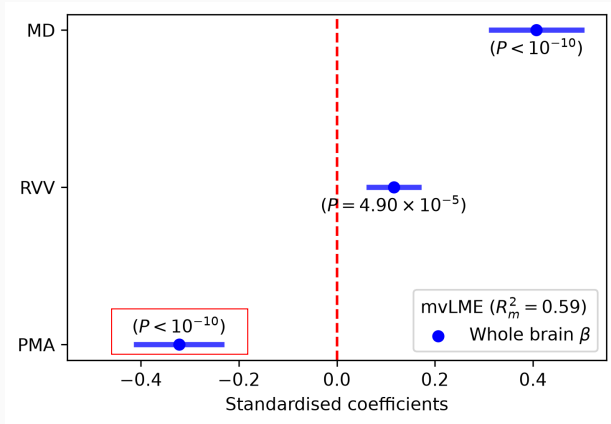
$$\text{Conductivity} \sim \text{PMA} + \text{MD} + \text{RVV}$$

Conductivity decline is not solely attributable to water content



- Modest RVV association is consistent with PVEs contamination
- **Developmental conductivity decline is a distinct effect** that is not solely explained by changes in water content.

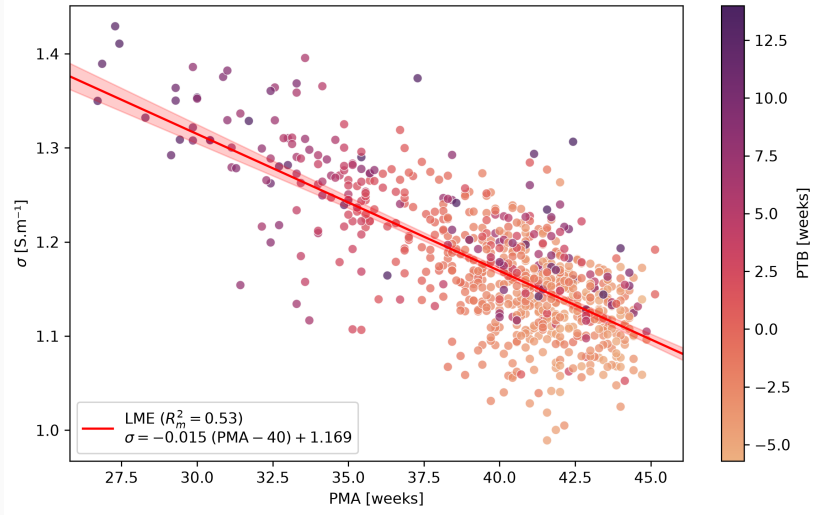
Conductivity decline is not solely attributable to water content



- Residual age-dependence is consistent with developmental changes in tissue ionic environment and membrane organisation.

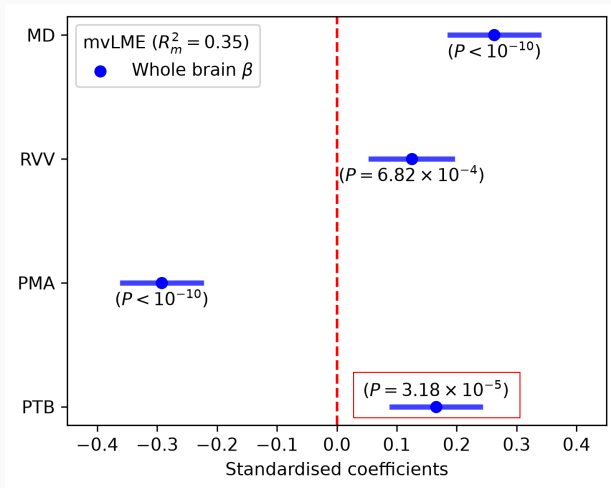
Sensitivity to prematurity

Modulation effect of prematurity



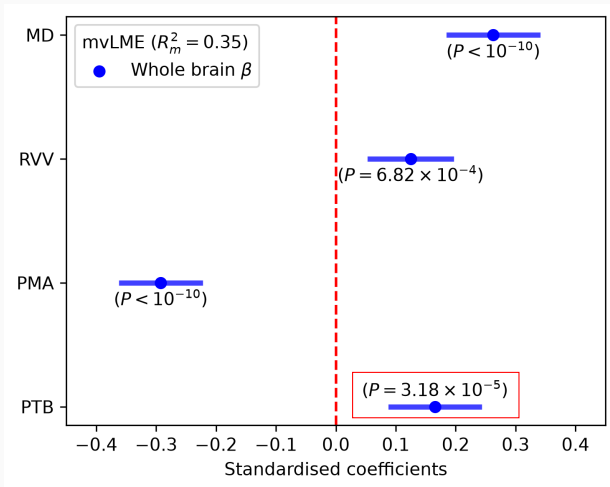
- Prematurity as **modulator** of the conductivity trajectory.
- Operationalised as continuous variable: $PTB = 37 - GA$.

Prematurity is associated with elevated conductivity



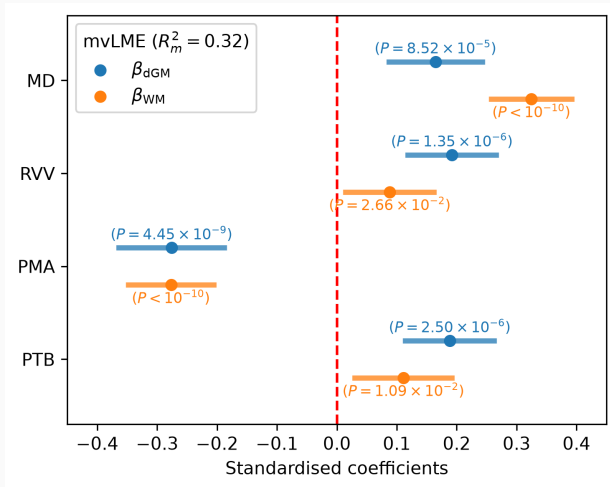
- Greater degree of prematurity was independently associated with **higher conductivity** at term-equivalent age.

Prematurity is associated with elevated conductivity



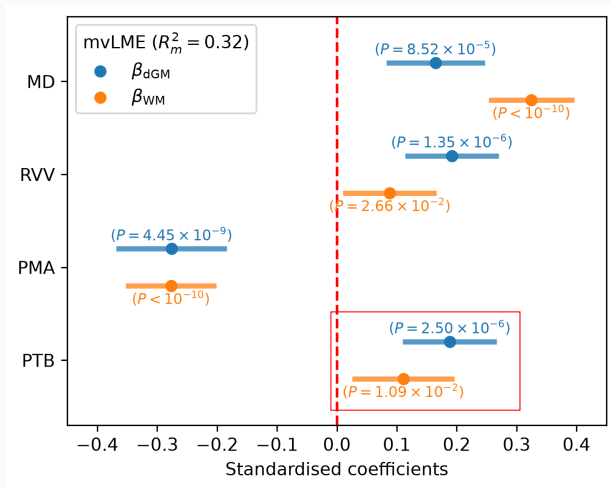
- Consistent with less mature tissue environment characterised by greater extracellular volume and lower membrane density.

Prematurity effect on conductivity is greater in dGM



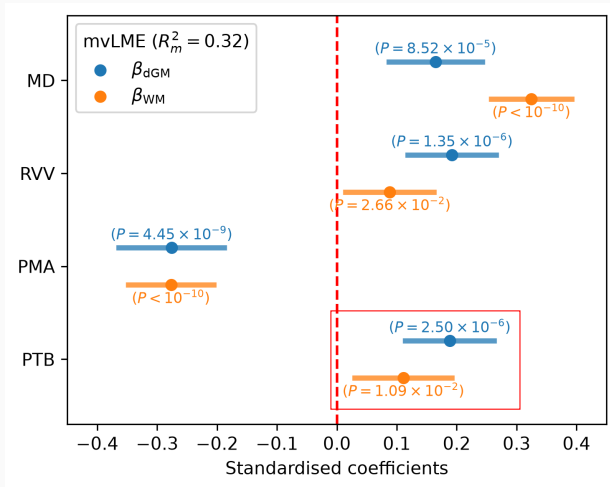
- Tissue-level modelling of deep grey matter (dGM) and white matter (WM) compartments.

Prematurity effect on conductivity is greater in dGM



- Stronger prematurity effect in dGM than WM (unstandardised Tissue $\times$ PTB interaction term, $P = 0.002$).

Prematurity effect on conductivity is greater in dGM

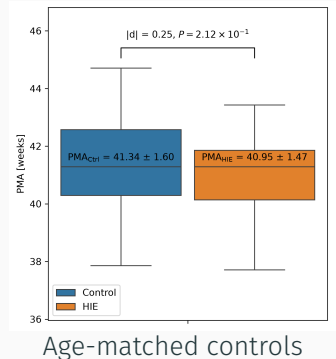


- Deep grey matter structures are recognised as especially vulnerable to prematurity-related disruption (Ball et al., 2013).

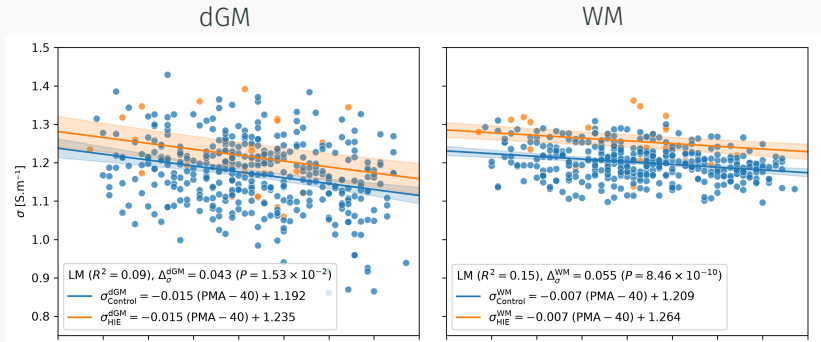
Brain injury model

HIE and control groups selection

- Hypoxic-ischemic encephalopathy (HIE) is expected to produce **ionic dysregulation** → changes in electrical conductivity.
- Term-age neonates with HIE compared to **age-matched controls**.
- Premature neonates excluded.

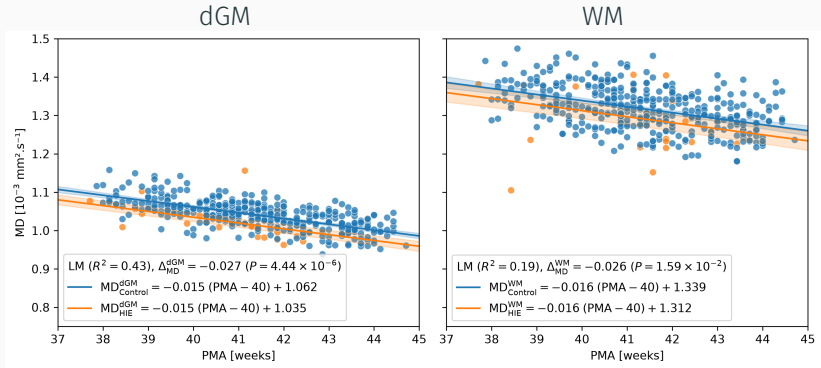


HIE is associated with elevated dGM and WM conductivity levels



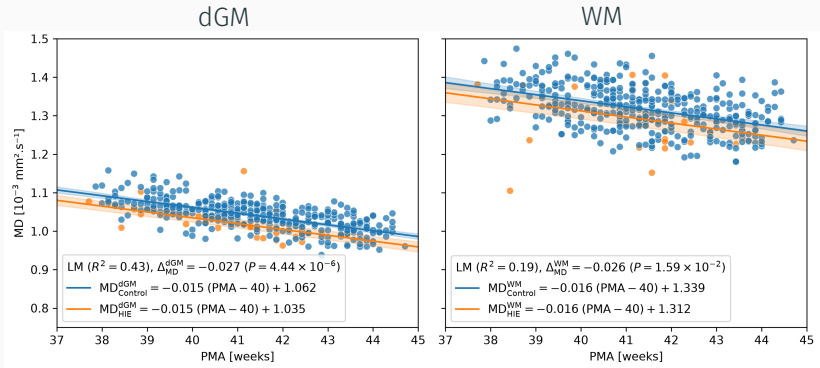
- Adjusting for PMA, **$\sim 5\%$ increase** in dGM and WM conductivity relative to term-born control.

Divergence of conductivity and diffusion measures following HIE



- MD reduction within the same tissue compartments.

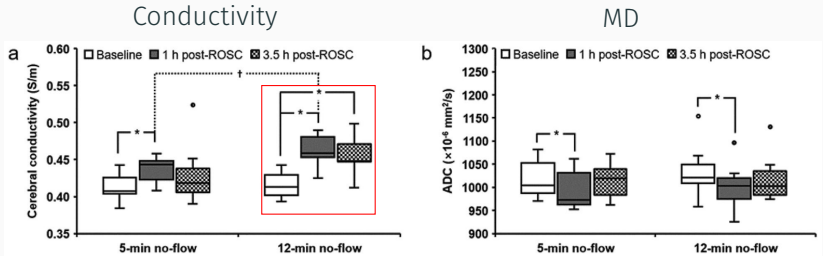
Divergence of conductivity and diffusion measures following HIE



- Conductivity elevation persisted after adjustment for MD and RVV in tissue-level multivariate models.
- **Genuine dissociation** between conductivity and MD measurements following HIE brain injury.

- **Distinct pathophysiological processes** revealed by:
 - **MD** → cytotoxic oedema **restricts extracellular water motion** (Bednarek et al., 2012).
 - **Conductivity** → alterations in cellular metabolism, extracellular metabolite concentrations, ion transporter function and pH regulation **disrupt ionic homeostasis** (Fleiss and Gressens, 2012, Helmy et al., 2012).

Comparison with HI animal model



Jung et al., 2024

- In an animal model, conductivity increased while diffusivity decreased following insult.
- Conductivity correlated with injury severity (Jung et al., 2024).

Acknowledgments

- Paper:
 - A. Boutillon, A. Price, R. Kurtzbard, O. Gale-Grant, E. Galanides, A. Lawrence, J. Hand, D. Batalle, J. O'Muircheartaigh, G. McAlonan, J. V. Hajnal, A. D. Edwards, T. Arichi, M. Deprez, and S. J. Malik, "Radiofrequency electrical conductivity reveals a distinct dimension of early human brain development," *Scientific Reports*, 2026. DOI: 10.1038/s41598-026-61290-3
- Code:
 - github.com/aboutill/nano-eptk
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